

Avneet Aurora

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Education: University of Delaware, Newark, DE, Honors College, B.S. Computer Engineering, GPA: 3.77, Dean's List

Relevant Coursework: Calculus III, Linear Algebra II, Data Structures, Intro to Software Engineering, Intro to Machine Learning

Languages: Python, JavaScript, TypeScript, C, Java, HTML, CSS, VHDL

Frameworks/Tools: React, NumPy, Pandas, PyTorch, Tensorflow, Bash, Git, Supabase, Vercel, AWS

Specializations: Full-Stack Development, Frontend Development, Machine Learning, API Design

Soft-Skills: Problem Solving, Critical Thinking, Adaptability, Verbal & Written Communication, Teamwork, Quick Learner

RELATED EXPERIENCE

University of Delaware , Computer Science— Vertically Integrated Projects, Newark, DE

Feb 2026 – Present

- Worked independently on extending a research paper to replace the BGK collision operator with an artificial neural network in the Lattice Boltzmann Method (LBM), scaling fluid simulations from 2D to 3D for complex physical modeling
- Implemented Python-based scientific computing modules with data structures optimized for large-scale numerical simulation and fluid dynamics prediction. Using libraries such as TensorFlow, NumPy, and Pandas.
- Applied debugging and problem-solving techniques to validate neural network collision operator outputs against established analytical solutions

University of Delaware, Electrical, Computer Engineering — Undergraduate Research, Newark, DE

July 2025– February 2026

- Replicating 3D wildfire estimation research using deep learning on ALS Point Cloud and 3D Tensor data; expanding methodology to satellite LiDAR analytics
- Coded backend data pipelines in Python script for point normalization, data generation, computing forest metrics to understand the structure of each forest or subplot, and to graph 2D plots for visualizations and heatmaps using Matplotlib.
- Built a Convolutional Neural Network with ReLu, Batch Normalization, and Regression layers in Pytorch to troubleshoot overfitting of data and early convergence.

Savio Institute – Teaching Assistant, Remote *August 2024 - December 2024*

- Helped a private tutoring company with rising 9th graders prepare for standardized tests, including SAT and Terranova.
- Assigned 2-3 students, taught high school-level Math and English, assigned homework, and used written communication skills.

Code Differently — Technology Intern, Buccini Pollin Group, Wilmington, DE

June 2024 – Aug 2024

- Performed financial analysis for real estate development projects; applied problem-solving and time management skills to present a data-driven proposal to BPG founders on amenity gym improvements
- Earned Microsoft Office, Excel, Word, and Digital Literacy certifications (IC3 Levels 1 & 2)
- Won a \$1,000 scholarship from the Metropolitan Urban League

PROJECT EXPERIENCE

Decorate Room AI — <https://www.decora.club>

Aug 2025 – Nov 2025

- Built a full-stack AI interior-design platform using React/Next.js + TypeScript; designed RESTful API endpoints (API design) for serverless backend deployment on Vercel. Learned agile methodologies in software engineering and pushed production-level code.
- Implemented backend development features, including Supabase Auth/DB/Storage integration and a Stripe-based credit system with object-oriented design patterns. Used SQL databases to store user information and placed images in buckets.
- Applied, tested, and analyzed data modeling for furniture detection pipelines and user session management; optimized user experience (UX) across the end-to-end redesign workflow

TaskAI — <https://devpost.com/software/taskai-vjx8cl>

Dec 2025

- Developed a Python web app for college students to schedule study sessions around classes and activities; integrated Google Gemini for intelligent scheduling recommendations
- Designed a frontend with a clean user interface (UI) in Drafter with a focus on user experience (UX); won Best Functional Implementation at the University of Delaware Honors Hackathon.

SquatBuddy — devpost.com/squatbuddy

March 2025

- Under biotech and healthcare, built an AI-powered powerlifting coach at HenHacks 2026 using an LSTM Autoencoder to analyze squat form frame-by-frame from raw video input.
- Applied deep learning and object-oriented design from OpenCV libraries to deliver joint-specific biomechanical feedback; engineered a real-time video processing pipeline across the full kinetic chain.

AWARDS

Winter Fellows Award, Dec 2025 – Selected for a \$400 merit scholarship for the project "Deep Learning for Forest Fingerprints" under Prof. Gonzalo Arce; presented findings at the Winter Undergraduate Research Showcase

Best Functional Implementation Award, Dec 2025 – TaskAI won for best implementation within Python and Drafter at Honors Hackathon

ACTIVITIES

DevLab Club — President, Newark, DE

Feb 2025 – Present

- Led and organized weekly sessions on full-stack and backend development topics, including Git, API design, Software Engineering Principles like object-oriented programming and use of LLMs like Gemini or ChatGPT; grew club to 20+ active members
- Mentor and collaborate with members, using verbal communication skills on outlining personal project design, problem-solving, and debugging code and architecture.